

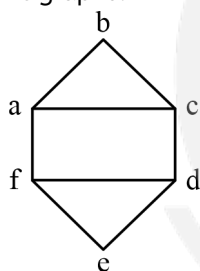
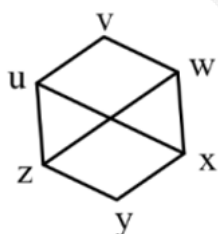
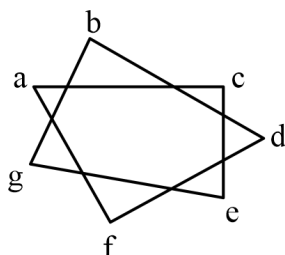
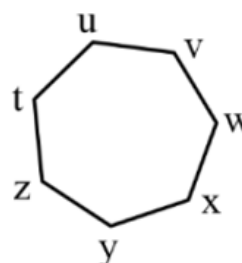
Computer Science & Information Technology

Discrete Mathematics

DPP: 2

Graph Theory

- Q1** Let G is simple graph with 7 vertices and 11 edges. Then find number of edges in complement of G
- Q2** If a self-complementary graph G is of size 18, then find the number of vertices in the graph G .
- Q3** If G is a simple graph with degree sequence $\{5, 2, 2, 2, 2, 1\}$ then what is the number of edges in the complement of G ?
- Q4** Which of the following options is/are correct for isomorphic graphs?

 G_1 : G_2 : G_3 : G_4 :

- (A) G_1 and G_2 are isomorphic graph.
- (B) G_3 and G_4 are isomorphic graph.
- (C) G_1 and G_2 are not isomorphic graph.
- (D) G_3 and G_4 are not isomorphic graph.

- Q5** How many simple non-isomorphic trees are possible with 5-vertices?
- Q6** Let G be a connected planar graph with 20 vertices each of degree 3 then the number of faces in the planar embedding of the graph is _____.
- Q7** Let G be a connected planar graph with 35 regions each of degree 6. Then the number of vertices in graph G is _____.
- Q8** Let G be a connected planar graph with 12 vertices and 30 edges, and degree of each region is k . Then the value of k is _____.
- Q9** Minimum number of vertices necessary in a simple connected planar graph with 11 edges is _____.
- Q10** Maximum number of regions possible in a simple connected planar graph with 10 vertices is _____.
- Q11** Consider a 5-regular connected planar graph with 10 vertices.
How many bounded faces are there in the planar embedding of the graph?



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(A) 15

(B) 16

(C) 17

(D) 18

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Answer Key

Q1 10~10
Q2 9~9
Q3 8~8
Q4 (B, C)
Q5 3~3
Q6 12~12

Q7 72~72
Q8 3~3
Q9 6~6
Q10 16~16
Q11 (B)

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Hints & Solutions

Q1 Text Solution:

Total number of edges possible in a simple graph with 7 vertices = ${}^7C_2 = 21$

Number of edges in complement of $G = 21 - 11 = 10$

Q2 Text Solution:

Let the number of vertices be n .

For any self complementary graph G ,

$$|E(G)| = |E(G')|$$

Also we know that for any simple graph,

$$|E(G)| + |E(G')| = {}^nC_2$$

$$\Rightarrow 18 + 18 = {}^nC_2$$

$$\Rightarrow 36 = {}^nC_2$$

$$\Rightarrow n = 9$$

Q3 Text Solution:

Total number of vertices in graph G is 6.

Total number of edges possible in a simple graph with 6 vertices = ${}^6C_2 = 15$

Total degree of the given graph = $5 + 2 + 2 + 2 + 2 + 1 = 14$

Number of edges in $G = 14/2 = 7$

Number of edges in the complement of $G = 15 - 7 = 8$

Q4 Text Solution:

G_1 and G_2 are not isomorphic as bijective mapping is not possible among the vertices.

Q5 Text Solution:

There are three simple non-isomorphic trees possible with 5-vertices.

Q6 Text Solution:

Total degree of the graph = no. of vertices $\times$

degree of each vertex = $20 \times 3 = 60$

Number of edges = total degree / 2 = $60 / 2 = 30$

$$v - e + f = 2$$

$$\Rightarrow 20 - 30 + f = 2$$

$$\Rightarrow f = 12$$

Q7 Text Solution:

Total degree = $35 \times 6 = 210$

Total edges = $210/2 = 105$

$$v - e + f = 2$$

$$\Rightarrow v - 105 + 35 = 2$$

$$\Rightarrow v = 72$$

Q8 Text Solution:

$$v - e + f = 2$$

$$\Rightarrow 12 - 30 + f = 2$$

$$\Rightarrow f = 20$$

We know that,

$$f \times k = 2 \times e$$

$$\Rightarrow 20 \times k = 2 \times 30$$

$$\Rightarrow k = 3$$

Q9 Text Solution:

In a simple connected planar graph,

$$e \leq 3v - 6$$

$$\Rightarrow 11 \leq 3v - 6$$

$$\Rightarrow 17 \leq 3v$$

$$\Rightarrow 5.66 \leq v$$

$$\Rightarrow v = 6$$

Q10 Text Solution:

$$e \leq 3v - 6$$

$$\Rightarrow e \leq 24$$

$$\Rightarrow e = 24$$

Also we know that,

$$3r \leq 2e$$

$$\Rightarrow 3r \leq 48$$

$$\Rightarrow r \leq 16$$

$$\Rightarrow r = 16$$

Q11 Text Solution:

Total degree = $5 \times 10 = 50$

$$e = 50/2 = 25$$

$$v - e + f = 2$$

$$10 - 25 + f = 2$$

$$\Rightarrow f = 17 = \text{total number of faces}$$

number of bounded faces = total number of

faces - total number of unbounded faces = $17 - 1 = 16$



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